Hugo Marquerie 04/11/2025

Propiedades de la convolución

Lema 1. *Sean $f, g, h\colon \mathbb{R}^n \to \mathbb{K} = \mathbb{R}$ o $\mathbb{C}$ dos funciones medibles, entonces*

(1) *Conmutatividad:* $f * g = g * f$ *c.t.p.*

(2) *Asociatividad:* $f * (g * h) = (f * g) * h$ *c.t.p.*

(3) $T_z\,(f * g) = (T_z f) * g = f * (T_z g)$ *c.t.p.*

(4) $\operatorname{supp}(f * g) \subset \operatorname{supp}(f) + \operatorname{supp}(g)$.

***Demostración*:**

(1) $f * g(x) = \displaystyle\int_{\mathbb{R}^n} f(x-y)\,g(y)\,\mathrm{d}y \overset{\substack{z=x-y\\ \downarrow}}{=} \int_{\mathbb{R}^n} f(z)\,g(x-z)\,\mathrm{d}z = g * f(x)$ c.t.p. $x \in \mathbb{R}^n$.

(2) Calculamos y aplicamos el Teorema de Fubini:

$$\begin{aligned}
(f * g) * h(x) &= \int_{\mathbb{R}^n} (f * g)\,(x-y)\,h(y)\,\mathrm{d}y = \int_{\mathbb{R}^n} g * f(x-y)\,h(y)\,\mathrm{d}y \\
&= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} g(x-y-z)\,f(z)\,\mathrm{d}z\right) h(y)\,\mathrm{d}y \\
&\overset{\substack{\text{Fubini}\\ \downarrow}}{=} \int_{\mathbb{R}^n} f(z) \left(\int_{\mathbb{R}^n} g(x-y-z)\,h(y)\,\mathrm{d}y\right) \mathrm{d}z = \int_{\mathbb{R}^n} g * h(x-z)\,f(z)\,\mathrm{d}z \\
&= f * (g * h)\,(x)
\end{aligned}$$

(3) Se demuestra la primera igualdad, la segunda es análoga. Tenemos que

$$\begin{aligned}
T_z\,(f * g)\,(x) = f * g(x-z) &= \int_{\mathbb{R}^n} f(x-z-y)\,g(y)\,\mathrm{d}y \\
&= \int_{\mathbb{R}^n} T_z f(x-y)\,g(y)\,\mathrm{d}y = (T_z f) * g(x).
\end{aligned}$$

(4) Basta probarlo para el soporte sin cerrar. Es decir, si $s(f) := \{x \in \mathbb{R}^n : f(x) \neq 0\}$ basta ver que $s(f * g) \subset s(f) + s(g)$. Sea $x \in \mathbb{R}^n \setminus (s(f) + s(g))$, entonces $\forall y \in s(g) :$ $f(x-y) = 0$ porque $f(x-y) \neq 0 \implies x - y \in s(f) \implies x = (x-y) + y \in s(f)$.

Como $f(x-y) = 0$, $f * g(x) = \int_{\mathbb{R}^n} f(x-y)\,g(y)\,\mathrm{d}y = 0$, luego $x \in \mathbb{R}^n \setminus s(f * g)$. ∎

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